

----- Evaluate basismodel_hh(1) -----
basismodel_hh(1)_states(5) --- loglike-Train: -41481.07750092586 /// loglike-Test: -44324.29438500175 /// MAE: 132.53100866540922 /// Mean CRPS: 4355.203181921675
basismodel_hh(1)_states(10) --- loglike-Train: -39065.950732290104 /// loglike-Test: -42194.5874940169 /// MAE: 120.10431027414074 /// Mean CRPS: 3982.880208333335
basismodel_hh(1)_states(15) --- loglike-Train: -38168.014405354465 /// loglike-Test: -41613.51812950947 /// MAE: 119.56503540442228 /// Mean CRPS: 3945.0950307377047
basismodel_hh(1)_states(20) --- loglike-Train: -37826.10300041805 /// loglike-Test: -41375.46839832357 /// MAE: 117.6750890979448 /// Mean CRPS: 3906.482382741348
basismodel_hh(1)_states(25) --- loglike-Train: -37373.48955201307 /// loglike-Test: -41184.45497966142 /// MAE: 118.98505532714763 /// Mean CRPS: 3924.761128187614
basismodel_hh(1)_states(30) --- loglike-Train: -36965.303381219805 /// loglike-Test: -40899.89986259395 /// MAE: 119.92534006151358 /// Mean CRPS: 3909.4012978142077
basismodel_hh(1)_states(40) --- loglike-Train: -36396.54648916758 /// loglike-Test: -40844.81619181107 /// MAE: 117.16873081870038 /// Mean CRPS: 3856.359460382516
basismodel_hh(1)_states(50) --- loglike-Train: -36756.72673189263 /// loglike-Test: -41146.59986969703 /// MAE: 118.11616528878334 /// Mean CRPS: 3876.2990095628415
basismodel_hh(1)_states(60) --- loglike-Train: -36044.44878474544 /// loglike-Test: -40672.80649253164 /// MAE: 117.35973543943837 /// Mean CRPS: 3869.5449112021856

----- Evaluate basismodel_hh(1) with timestamps -----
basismodel_ts(4)_states(5) --- loglike-Train: -52029.696811524125 /// loglike-Test: -56011.150468617154 /// MAE: 146.88093854675395 /// Mean CRPS: 4831.758054417122
basismodel_ts(4)_states(10) --- loglike-Train: -55815.00152774333 /// loglike-Test: -58005.933133924904 /// MAE: 128.35463907857138 /// Mean CRPS: 4289.545736566484
basismodel_ts(4)_states(15) --- loglike-Train: -46062.47744723051 /// loglike-Test: -49828.35348881935 /// MAE: 126.18851326680549 /// Mean CRPS: 4122.861509562841
basismodel_ts(4)_states(20) --- loglike-Train: -45167.54519487994 /// loglike-Test: -48936.835176307184 /// MAE: 122.10108364432337 /// Mean CRPS: 4019.9763775045535
basismodel_ts(4)_states(25) --- loglike-Train: -44977.736741228466 /// loglike-Test: -49027.771726336796 /// MAE: 124.18253831966923 /// Mean CRPS: 4064.048924180328
basismodel_ts(4)_states(30) --- loglike-Train: -44329.92660914041 /// loglike-Test: -48277.186887119744 /// MAE: 121.6075483476203 /// Mean CRPS: 3987.20864640255
basismodel_ts(4)_states(40) --- loglike-Train: -43840.35680403838 /// loglike-Test: -48147.01641639687 /// MAE: 121.58618391836202 /// Mean CRPS: 3993.2903859289618
basismodel_ts(4)_states(50) --- loglike-Train: -42851.575881576355 /// loglike-Test: -47717.5220379589 /// MAE: 119.88250215467117 /// Mean CRPS: 3940.4084130236793
basismodel_ts(4)_states(60) --- loglike-Train: -42778.13313464001 /// loglike-Test: -48050.09702177362 /// MAE: 121.35522928108195 /// Mean CRPS: 3975.1909437613845

basismodel_ts(8)_states(5) --- loglike-Train: -82970.40141472811 /// loglike-Test: -87195.17351303354 /// MAE: 148.10202247133117 /// Mean CRPS: 5144.138518897997
basismodel_ts(8)_states(10) --- loglike-Train: -55815.00152774333 /// loglike-Test: -59999.37218065791 /// MAE: 150.34195921212873 /// Mean CRPS: 4957.5914162121293
basismodel_ts(8)_states(15) --- loglike-Train: -52603.1279585926 /// loglike-Test: -63973.308595885566 /// MAE: 140.65499748289676 /// Mean CRPS: 4741.653973132969
basismodel_ts(8)_states(20) --- loglike-Train: -53238.859122695074 /// loglike-Test: -57481.366793468056 /// MAE: 142.58021499691077 /// Mean CRPS: 4632.710410974499
basismodel_ts(8)_states(25) --- loglike-Train: -50539.12544211864 /// loglike-Test: -55139.36975903339 /// MAE: 131.92828113675915 /// Mean CRPS: 4275.666268214936
basismodel_ts(8)_states(30) --- loglike-Train: -50516.15385994074 /// loglike-Test: -55260.107683002027 /// MAE: 128.86373534674786 /// Mean CRPS: 4279.855390482696
basismodel_ts(8)_states(40) --- loglike-Train: -48723.1276400675 /// loglike-Test: -53652.3625527995 /// MAE: 129.53863971659908 /// Mean CRPS: 4169.328483069217
basismodel_ts(8)_states(50) --- loglike-Train: -48279.19166065764 /// loglike-Test: -53520.07187172862 /// MAE: 127.19606025655628 /// Mean CRPS: 4144.617970173042
basismodel_ts(8)_states(60) --- loglike-Train: -47606.58243669174 /// loglike-Test: -53445.999400827124 /// MAE: 126.65541538565827 /// Mean CRPS: 4115.640311930783

basismodel_ts(12)_states(5) --- loglike-Train: -97501.3707529801 /// loglike-Test: -103429.33211496277 /// MAE: 145.20347820154925 /// Mean CRPS: 5182.936991484517
basismodel_ts(12)_states(10) --- loglike-Train: -73159.84933934698 /// loglike-Test: -78843.6025214327 /// MAE: 149.87296415845185 /// Mean CRPS: 4930.63117001822
basismodel_ts(12)_states(15) --- loglike-Train: -58711.53213338714 /// loglike-Test: -64356.12879086158 /// MAE: 149.5489689568153 /// Mean CRPS: 4898.675973360656
basismodel_ts(12)_states(20) --- loglike-Train: -59543.28310651604 /// loglike-Test: -65464.27414707037 /// MAE: 145.2871481918089 /// Mean CRPS: 4655.572859744991
basismodel_ts(12)_states(25) --- loglike-Train: -55505.369886390545 /// loglike-Test: -61267.496499277826 /// MAE: 141.02405721651124 /// Mean CRPS: 4616.453381147541
basismodel_ts(12)_states(30) --- loglike-Train: -58206.18294495526 /// loglike-Test: -62760.052096840096 /// MAE: 135.49945198784232 /// Mean CRPS: 4442.079234972678
basismodel_ts(12)_states(40) --- loglike-Train: -53095.563176912685 /// loglike-Test: -59252.55752249936 /// MAE: 136.90665677083877 /// Mean CRPS: 4393.492173269581
basismodel_ts(12)_states(50) --- loglike-Train: -52565.277626941366 /// loglike-Test: -59434.470484524805 /// MAE: 128.87209070827092 /// Mean CRPS: 4245.34591302368
basismodel_ts(12)_states(60) --- loglike-Train: -51427.884708957325 /// loglike-Test: -58574.63428480178 /// MAE: 127.43934108198776 /// Mean CRPS: 4161.979650500911

basismodel_ts(24)_states(5) --- loglike-Train: -125740.43916832459 /// loglike-Test: -133572.01591223298 /// MAE: 148.39456998585624 /// Mean CRPS: 5235.255066029144
basismodel_ts(24)_states(10) --- loglike-Train: -96548.91105207108 /// loglike-Test: -104080.50829039999 /// MAE: 147.55521388565128 /// Mean CRPS: 4978.484744990893
basismodel_ts(24)_states(15) --- loglike-Train: -85517.22080369649 /// loglike-Test: -93685.20700967181 /// MAE: 151.7854375957615 /// Mean CRPS: 5107.824168943534
basismodel_ts(24)_states(20) --- loglike-Train: -77710.43496827612 /// loglike-Test: -85315.16024364065 /// MAE: 149.29939808999012 /// Mean CRPS: 4980.859403460838
basismodel_ts(24)_states(25) --- loglike-Train: -69100.75021611206 /// loglike-Test: -77156.1492964977 /// MAE: 148.46392902612521 /// Mean CRPS: 4885.235228825137
basismodel_ts(24)_states(30) --- loglike-Train: -66867.98309915277 /// loglike-Test: -76102.87267116416 /// MAE: 150.15844021576166 /// Mean CRPS: 4917.851633652094
basismodel_ts(24)_states(40) --- loglike-Train: -65912.07445103953 /// loglike-Test: -76222.99109337296 /// MAE: 143.01522789305395 /// Mean CRPS: 4690.4390368852455
basismodel_ts(24)_states(50) --- loglike-Train: -60608.94129372001 /// loglike-Test: -72496.82120776063 /// MAE: 139.07657407916426 /// Mean CRPS: 4647.465021630237
basismodel_ts(24)_states(60) --- loglike-Train: -59473.5122068869 /// loglike-Test: -70367.21933337505 /// MAE: 143.81640299671525 /// Mean CRPS: 4669.259961293261

basismodel_ts(96)_states(5) --- loglike-Train: -163329.17815060992 /// loglike-Test: -182927.35078793005 /// MAE: 150.1566210867106 /// Mean CRPS: 5258.604109744991
basismodel_ts(96)_states(10) --- loglike-Train: -149066.98454634118 /// loglike-Test: -170254.7397445254 /// MAE: 146.97199638386667 /// Mean CRPS: 5191.194045992714
basismodel_ts(96)_states(15) --- loglike-Train: -127411.45923532853 /// loglike-Test: -148295.8269407786 /// MAE: 149.24327970131418 /// Mean CRPS: 5245.2105248178505
basismodel_ts(96)_states(20) --- loglike-Train: -125626.41831404406 /// loglike-Test: -146928.33961500804 /// MAE: 148.33107970804718 /// Mean CRPS: 5211.331028005465
basismodel_ts(96)_states(25) --- loglike-Train: -114249.17045291455 /// loglike-Test: -135339.82750537162 /// MAE: 148.90418862841562 /// Mean CRPS: 5235.838883196721
basismodel_ts(96)_states(30) --- loglike-Train: -112005.32244600962 /// loglike-Test: -134480.9734796178 /// MAE: 149.79307415513645 /// Mean CRPS: 5252.400842440801
basismodel_ts(96)_states(40) --- loglike-Train: -93181.65574402298 /// loglike-Test: -114262.2403251336 /// MAE: 150.14095118338972 /// Mean CRPS: 5183.308145491803
basismodel_ts(96)_states(50) --- loglike-Train: -90396.97133061211 /// loglike-Test: -112153.38321951356 /// MAE: 150.00573212715608 /// Mean CRPS: 5220.244933970856
basismodel_ts(96)_states(60) --- loglike-Train: -78975.75090150666 /// loglike-Test: -100320.09069297508 /// MAE: 148.4349401515368 /// Mean CRPS: 5117.952299635701

Timing: [5763.77]sec

----- Evaluate seasonmodel_hh(1) -----
seasonmodel_(1)_spring_states(5) --- loglike-Train: -10442.388909770292 /// loglike-Test: -11168.521318341864 /// MAE: 137.8940809283159 /// Mean CRPS: 4477.877377717391
seasonmodel_(1)_spring_states(10) --- loglike-Train: -9199.268435289481 /// loglike-Test: -10325.885478894792 /// MAE: 123.73270685891173 /// Mean CRPS: 4086.402287137681
seasonmodel_(1)_spring_states(15) --- loglike-Train: -8931.401996199 /// loglike-Test: -10339.297526638438 /// MAE: 124.67790888020328 /// Mean CRPS: 4134.211050724638
seasonmodel_(1)_spring_states(20) --- loglike-Train: -8787.505406896704 /// loglike-Test: -10621.648179814893 /// MAE: 124.6102174354318 /// Mean CRPS: 4148.306725543478
seasonmodel_(1)_spring_states(25) --- loglike-Train: -8777.719486568134 /// loglike-Test: -10568.610974869633 /// MAE: 128.6799646467881 /// Mean CRPS: 4197.483242753623
seasonmodel_(1)_spring_states(30) --- loglike-Train: -8601.387236273325 /// loglike-Test: -10750.912226899825 /// MAE: 124.45168120776673 /// Mean CRPS: 4174.010416666667
seasonmodel_(1)_spring_states(40) --- loglike-Train: -8441.625878325478 /// loglike-Test: -10781.09433295496 /// MAE: 126.12386362398317 /// Mean CRPS: 4198.67536231884
seasonmodel_(1)_spring_states(50) --- loglike-Train: -8304.700503999893 /// loglike-Test: -11094.01204692912 /// MAE: 125.85378697876223 /// Mean CRPS: 4213.958786231884
seasonmodel_(1)_spring_states(60) --- loglike-Train: -8160.940904676856 /// loglike-Test: -11154.674481565078 /// MAE: 127.34740608130849 /// Mean CRPS: 4270.751245471014

seasonmodel_(1)_summer_states(5) --- loglike-Train: -10241.36701938602 /// loglike-Test: -11183.025534358167 /// MAE: 113.99353868534182 /// Mean CRPS: 3906.9033061594205
seasonmodel_(1)_summer_states(10) --- loglike-Train: -9860.396882726665 /// loglike-Test: -10829.047340267241 /// MAE: 113.76129673540746 /// Mean CRPS: 3836.537930253623
seasonmodel_(1)_summer_states(15) --- loglike-Train: -9646.046454353323 /// loglike-Test: -10786.166549663976 /// MAE: 113.28037328754516 /// Mean CRPS: 3794.569972826087
seasonmodel_(1)_summer_states(20) --- loglike-Train: -9539.14042812945 /// loglike-Test: -10847.311198715892 /// MAE: 111.59854481575333 /// Mean CRPS: 3745.861865942029
seasonmodel_(1)_summer_states(25) --- loglike-Train: -9612.430995377723 /// loglike-Test: -10827.032942956324 /// MAE: 115.32176784083606 /// Mean CRPS: 3802.6106204710145
seasonmodel_(1)_summer_states(30) --- loglike-Train: -9459.475926873807 /// loglike-Test: -10816.906181230508 /// MAE: 115.49323977565936 /// Mean CRPS: 3819.1064311594205
seasonmodel_(1)_summer_states(40) --- loglike-Train: -9066.351322411534 /// loglike-Test: -10898.781958053663 /// MAE: 120.71262314826613 /// Mean CRPS: 3862.180819746377
seasonmodel_(1)_summer_states(50) --- loglike-Train: -8862.56767943457 /// loglike-Test: -1068.878591754292 /// MAE: 115.65077939307282 /// Mean CRPS: 3831.748301630435
seasonmodel_(1)_summer_states(60) --- loglike-Train: -8889.79661063261 /// loglike-Test: -10972.929001580298 /// MAE: 118.25739547674142 /// Mean CRPS: 3836.074954710145

seasonmodel_(1)_fall_states(5) --- loglike-Train: -10181.48471690847 /// loglike-Test: -10803.586946079373 /// MAE: 123.63359708224436 /// Mean CRPS: 4044.043727106227
seasonmodel_(1)_fall_states(10) --- loglike-Train: -9332.857960643443 /// loglike-Test: -10235.209389790225 /// MAE: 116.92825926393542 /// Mean CRPS: 3786.890911721614
seasonmodel_(1)_fall_states(15) --- loglike-Train: -9097.128341556205 /// loglike-Test: -10180.48727727213 /// MAE: 112.22026127969269 /// Mean CRPS: 3684.872367216173
seasonmodel_(1)_fall_states(20) --- loglike-Train: -8833.410915252956 /// loglike-Test: -10149.544208262634 /// MAE: 112.9303101347219 /// Mean CRPS: 3720.3948031135533
seasonmodel_(1)_fall_states(25) --- loglike-Train: -8863.903989267035 /// loglike-Test: -10242.277867412575 /// MAE: 116.1567957157905 /// Mean CRPS: 3767.646273992675
seasonmodel_(1)_fall_states(30) --- loglike-Train: -8863.231747743714 /// loglike-Test: -10201.589123930793 /// MAE: 114.72024508646518 /// Mean CRPS: 3761.8305860805863
seasonmodel_(1)_fall_states(40) --- loglike-Train: -8435.467358927264 /// loglike-Test: -10327.577056353944 /// MAE: 114.4152857657062 /// Mean CRPS: 3721.2706043956046
seasonmodel_(1)_fall_states(50) --- loglike-Train: -8345.325088033955 /// loglike-Test: -10650.840552183296 /// MAE: 116.76309279966891 /// Mean CRPS: 3804.142513736264
seasonmodel_(1)_fall_states(60) --- loglike-Train: -8197.861981090751 /// loglike-Test: -10764.702054766873 /// MAE: 115.40567973495294 /// Mean CRPS: 3806.8150183150183

seasonmodel_(1)_winter_states(5) --- loglike-Train: -9646.756711564074 /// loglike-Test: -10715.520258072222 /// MAE: 142.01913322258346 /// Mean CRPS: 4745.630723443223
seasonmodel_(1)_winter_states(10) --- loglike-Train: -8863.533497579265 /// loglike-Test: -10284.964906156347 /// MAE: 129.17916426180625 /// Mean CRPS: 4311.651556776556
seasonmodel_(1)_winter_states(15) --- loglike-Train: -8594.419701658459 /// loglike-Test: -10213.90903844691 /// MAE: 126.50466370689183 /// Mean CRPS: 4270.486721611722
seasonmodel_(1)_winter_states(20) --- loglike-Train: -8433.381539277118 /// loglike-Test: -10353.793521174111 /// MAE: 131.41431386121698 /// Mean CRPS: 4245.731913919414
seasonmodel_(1)_winter_states(25) --- loglike-Train: -8400.464635935903 /// loglike-Test: -10344.898095162023 /// MAE: 128.5848656293241 /// Mean CRPS: 4295.8801510989015
seasonmodel_(1)_winter_states(30) --- loglike-Train: -8265.359416523996 /// loglike-Test: -10355.322437932466 /// MAE: 133.61941512362762 /// Mean CRPS: 4389.836652930403
seasonmodel_(1)_winter_states(40) --- loglike-Train: -8027.017221160989 /// loglike-Test: -10810.209719351817 /// MAE: 130.09092009744288 /// Mean CRPS: 4471.5294184981685
seasonmodel_(1)_winter_states(50) --- loglike-Train: -7993.302748923456 /// loglike-Test: -10767.40378283144 /// MAE: 127.65149678246603 /// Mean CRPS: 4386.447802197802
seasonmodel_(1)_winter_states(60) --- loglike-Train: -7875.145118647594 /// loglike-Test: -11077.57834183559 /// MAE: 130.9435592430973 /// Mean CRPS: 4412.86687710622

Timing: [305.114]sec

----- Evaluate basismodel_hh(2) -----

basismodel_hh(2)_states(5) --- loglike-Train: -46470.11171640476 /// loglike-Test: -51888.492152435356 /// MAE: 129.65215316456113 /// Mean CRPS: 4623.897370218579
basismodel_hh(2)_states(10) --- loglike-Train: -41214.2895935932 /// loglike-Test: -48459.75442281592 /// MAE: 121.5603385539863 /// Mean CRPS: 4210.90098744991
basismodel_hh(2)_states(15) --- loglike-Train: -40270.14560369384 /// loglike-Test: -47540.536959141056 /// MAE: 115.50879243997197 /// Mean CRPS: 4002.266478825137
basismodel_hh(2)_states(20) --- loglike-Train: -40088.368721428815 /// loglike-Test: -47359.81270074565 /// MAE: 118.15495256128234 /// Mean CRPS: 4035.2564606102005
basismodel_hh(2)_states(25) --- loglike-Train: -40153.26235251373 /// loglike-Test: -47073.20152336066 /// MAE: 118.01840572230851 /// Mean CRPS: 4023.2332650273224
basismodel_hh(2)_states(30) --- loglike-Train: -39945.2474362635 /// loglike-Test: -47236.69700634177 /// MAE: 117.67615485982276 /// Mean CRPS: 4008.2501423041895
basismodel_hh(2)_states(40) --- loglike-Train: -39739.03977520795 /// loglike-Test: -46889.87888366108 /// MAE: 116.33680582558142 /// Mean CRPS: 3976.9169228142077
basismodel_hh(2)_states(50) --- loglike-Train: -39622.118715858756 /// loglike-Test: -47336.15843209529 /// MAE: 116.2950344684441 /// Mean CRPS: 3990.2564606102005
basismodel_hh(2)_states(60) --- loglike-Train: -39560.084636168154 /// loglike-Test: -47338.3077043504 /// MAE: 116.0618213607454 /// Mean CRPS: 3980.917862021858

----- Evaluate basismodel_hh(2) with timestamps -----

basismodel_ts(4)_states(5) --- loglike-Train: -65535.32062746103 /// loglike-Test: -67940.45689479147 /// MAE: 151.39854474893707 /// Mean CRPS: 5644.551087204008
basismodel_ts(4)_states(10) --- loglike-Train: -53272.04930828631 /// loglike-Test: -59907.20020248825 /// MAE: 137.7260689525231 /// Mean CRPS: 4790.304844034608
basismodel_ts(4)_states(15) --- loglike-Train: -51001.704652076456 /// loglike-Test: -58350.551649990564 /// MAE: 130.27532393282206 /// Mean CRPS: 4637.995104735884
basismodel_ts(4)_states(20) --- loglike-Train: -48620.53202683863 /// loglike-Test: -56950.71686608286 /// MAE: 133.93814546421433 /// Mean CRPS: 4611.668317395264
basismodel_ts(4)_states(25) --- loglike-Train: -47191.17632395927 /// loglike-Test: -55600.15373919824 /// MAE: 125.0293322463008 /// Mean CRPS: 4351.852060564663
basismodel_ts(4)_states(30) --- loglike-Train: -47344.31885879664 /// loglike-Test: -55791.23785335908 /// MAE: 122.5326984453572 /// Mean CRPS: 4297.703779599271
basismodel_ts(4)_states(40) --- loglike-Train: -45198.04282087055 /// loglike-Test: -54233.1178219827 /// MAE: 118.67146496067313 /// Mean CRPS: 4109.714537795992
basismodel_ts(4)_states(50) --- loglike-Train: -44878.63515337054 /// loglike-Test: -54782.09095906868 /// MAE: 118.09081262390774 /// Mean CRPS: 4126.662283697632
basismodel_ts(4)_states(60) --- loglike-Train: -44565.602809644435 /// loglike-Test: -54543.00231414288 /// MAE: 118.34122071769643 /// Mean CRPS: 4137.684304086995

basismodel_ts(8)_states(5) --- loglike-Train: -96383.9588452086 /// loglike-Test: -99298.25670271984 /// MAE: 157.22679704516744 /// Mean CRPS: 5795.702698087432
basismodel_ts(8)_states(10) --- loglike-Train: -65059.623536829626 /// loglike-Test: -69387.14018107904 /// MAE: 151.52155032406426 /// Mean CRPS: 5384.729565118397
basismodel_ts(8)_states(15) --- loglike-Train: -63548.2996658004 /// loglike-Test: -67661.64363526081 /// MAE: 144.9391867147984 /// Mean CRPS: 5235.355561247223
basismodel_ts(8)_states(20) --- loglike-Train: -65754.173912704486 /// loglike-Test: -64983.93750222899 /// MAE: 137.9004071892822 /// Mean CRPS: 4734.01024590164
basismodel_ts(8)_states(25) --- loglike-Train: -55343.95587123324 /// loglike-Test: -63362.88053116272 /// MAE: 135.30096989939847 /// Mean CRPS: 4667.338285519126
basismodel_ts(8)_states(30) --- loglike-Train: -55327.724888100194 /// loglike-Test: -62502.66347168328 /// MAE: 130.57423217907416 /// Mean CRPS: 4604.462716302368
basismodel_ts(8)_states(40) --- loglike-Train: -51027.07880559975 /// loglike-Test: -60440.344759758125 /// MAE: 124.43677087996679 /// Mean CRPS: 4338.087801684882
basismodel_ts(8)_states(50) --- loglike-Train: -50786.32456150908 /// loglike-Test: -60436.59902951782 /// MAE: 121.82781362224587 /// Mean CRPS: 4285.048952641166
basismodel_ts(8)_states(60) --- loglike-Train: -50208.53375212065 /// loglike-Test: -59611.97785959323 /// MAE: 118.62716490099963 /// Mean CRPS: 4158.2545821949

basismodel_ts(12)_states(5) --- loglike-Train: -101947.9924843064 /// loglike-Test: -105517.5807044371 /// MAE: 151.01519292656178 /// Mean CRPS: 5559.858407331511
basismodel_ts(12)_states(10) --- loglike-Train: -80358.34516544727 /// loglike-Test: -85021.52029995453 /// MAE: 152.32367785721877 /// Mean CRPS: 5537.793118169399
basismodel_ts(12)_states(15) --- loglike-Train: -69070.03400332708 /// loglike-Test: -73139.92176728741 /// MAE: 150.45692783209745 /// Mean CRPS: 5440.413820582878
basismodel_ts(12)_states(20) --- loglike-Train: -65754.173912704486 /// loglike-Test: -71914.67170583646 /// MAE: 145.80651459231063 /// Mean CRPS: 5224.281193078325
basismodel_ts(12)_states(25) --- loglike-Train: -65729.22440324965 /// loglike-Test: -72724.22088263843 /// MAE: 142.45402837366643 /// Mean CRPS: 4945.3278973132965
basismodel_ts(12)_states(30) --- loglike-Train: -59636.765318171136 /// loglike-Test: -67053.32726606319 /// MAE: 133.42598032771608 /// Mean CRPS: 4683.4935393897995
basismodel_ts(12)_states(40) --- loglike-Train: -56604.9534769891 /// loglike-Test: -66233.32086689363 /// MAE: 130.10681927686423 /// Mean CRPS: 4589.621442395264
basismodel_ts(12)_states(50) --- loglike-Train: -56674.25511346693 /// loglike-Test: -66513.8036651926 /// MAE: 134.95461699796277 /// Mean CRPS: 4676.113444899818
basismodel_ts(12)_states(60) --- loglike-Train: -53672.053815649924 /// loglike-Test: -64805.79138258647 /// MAE: 127.94000013614288 /// Mean CRPS: 4469.8748576958105
basismodel_ts(24)_states(40) --- loglike-Train: -72936.2337597912 /// loglike-Test: -80240.5937507594 /// MAE: 142.6158454742262 /// Mean CRPS: 5143.656278460838
basismodel_ts(24)_states(50) --- loglike-Train: -70193.49146319901 /// loglike-Test: -81384.17328773272 /// MAE: 139.05975356186588 /// Mean CRPS: 4969.788564435337
basismodel_ts(24)_states(60) --- loglike-Train: -65721.19192987318 /// loglike-Test: -77605.335232321 /// MAE: 138.80044622680327 /// Mean CRPS: 4934.986338797814

basismodel_ts(24)_states(5) --- loglike-Train: -129369.76830497193 /// loglike-Test: -135609.29876953387 /// MAE: 152.04575280720192 /// Mean CRPS: 5754.466160063752
basismodel_ts(24)_states(10) --- loglike-Train: -101400.44468681741 /// loglike-Test: -106919.69650370287 /// MAE: 153.58737804166452 /// Mean CRPS: 5535.988985655737
basismodel_ts(24)_states(15) --- loglike-Train: -93896.71756609056 /// loglike-Test: -98474.95688222058 /// MAE: 150.8588598455939 /// Mean CRPS: 5623.049066484517
basismodel_ts(24)_states(20) --- loglike-Train: -88637.43060898237 /// loglike-Test: -94994.36437512128 /// MAE: 151.30094179116088 /// Mean CRPS: 5580.194045992714
basismodel_ts(24)_states(25) --- loglike-Train: -80248.38443018249 /// loglike-Test: -86762.34949856745 /// MAE: 147.72285759545485 /// Mean CRPS: 5423.6957251821495
basismodel_ts(24)_states(30) --- loglike-Train: -75707.5552692642 /// loglike-Test: -82159.65483399534 /// MAE: 148.69204985706375 /// Mean CRPS: 5386.50587449909
basismodel_ts(24)_states(40) --- loglike-Train: -72936.2337597912 /// loglike-Test: -80240.5937507594 /// MAE: 142.6158454742262 /// Mean CRPS: 5143.656278460838
basismodel_ts(24)_states(50) --- loglike-Train: -70193.49146319901 /// loglike-Test: -81384.17328773272 /// MAE: 139.05975356186588 /// Mean CRPS: 4969.788564435337
basismodel_ts(24)_states(60) --- loglike-Train: -65721.19192987318 /// loglike-Test: -77605.335232321 /// MAE: 138.80044622680327 /// Mean CRPS: 4934.986338797814

basismodel_ts(96)_states(5) --- loglike-Train: -176526.3973998261 /// loglike-Test: -195652.52462032228 /// MAE: 159.6883017670856 /// Mean CRPS: 5918.600836748634
basismodel_ts(96)_states(10) --- loglike-Train: -154345.43520709645 /// loglike-Test: -174427.5097979699 /// MAE: 158.38607408023643 /// Mean CRPS: 5901.892615346038
basismodel_ts(96)_states(15) --- loglike-Train: -137578.7810051721 /// loglike-Test: -158057.84462693782 /// MAE: 155.7691899863659 /// Mean CRPS: 5840.706454918033
basismodel_ts(96)_states(20) --- loglike-Train: -133866.50912040117 /// loglike-Test: -154651.23587630194 /// MAE: 159.41723909360053 /// Mean CRPS: 5907.106642759562
basismodel_ts(96)_states(25) --- loglike-Train: -123802.98090007187 /// loglike-Test: -143727.48820623156 /// MAE: 152.88998951059241 /// Mean CRPS: 5757.1880122958054
basismodel_ts(96)_states(30) --- loglike-Train: -117468.43150132653 /// loglike-Test: -138955.07891163265 /// MAE: 152.81285947301 /// Mean CRPS: 5786.322176684882
basismodel_ts(96)_states(40) --- loglike-Train: -102672.49017528095 /// loglike-Test: -121999.85863259382 /// MAE: 157.56051744928305 /// Mean CRPS: 5830.626764571949
basismodel_ts(96)_states(50) --- loglike-Train: -96655.70522707945 /// loglike-Test: -118598.19674451319 /// MAE: 152.6084766251461 /// Mean CRPS: 5691.190317622951
basismodel_ts(96)_states(60) --- loglike-Train: -91796.59642409423 /// loglike-Test: -112561.18374526131 /// MAE: 154.06217302335395 /// Mean CRPS: 5795.064691484517

Timing: [6757.337]sec

----- Evaluate seasonmodel_hh(2) -----

seasonmodel_(2)_spring_states(5) --- loglike-Train: -11765.152227269875 /// loglike-Test: -12014.972043602844 /// MAE: 127.66763586254018 /// Mean CRPS: 4303.703917572464
seasonmodel_(2)_spring_states(10) --- loglike-Train: -11157.663454384361 /// loglike-Test: -12452.52632283612 /// MAE: 115.111552812429 /// Mean CRPS: 3936.7502264492755
seasonmodel_(2)_spring_states(15) --- loglike-Train: -11026.533084195087 /// loglike-Test: -11488.508705491276 /// MAE: 114.92580120857815 /// Mean CRPS: 3940.236526268116
seasonmodel_(2)_spring_states(20) --- loglike-Train: -11030.312937507233 /// loglike-Test: -11477.775545254146 /// MAE: 115.92551090494169 /// Mean CRPS: 3959.246037137681
seasonmodel_(2)_spring_states(25) --- loglike-Train: -10936.61024879303 /// loglike-Test: -11515.653673269204 /// MAE: 114.6940077265713 /// Mean CRPS: 3938.8094429347825
seasonmodel_(2)_spring_states(30) --- loglike-Train: -10967.604553732317 /// loglike-Test: -11524.445988553878 /// MAE: 116.44339135702307 /// Mean CRPS: 3965.3410326086955
seasonmodel_(2)_spring_states(40) --- loglike-Train: -10746.758901830686 /// loglike-Test: -11665.027114205554 /// MAE: 116.4972838383979 /// Mean CRPS: 3984.452332427536
seasonmodel_(2)_spring_states(50) --- loglike-Train: -10668.410730518299 /// loglike-Test: -11793.52196608514 /// MAE: 115.96047792840218 /// Mean CRPS: 3990.5707653985055
seasonmodel_(2)_spring_states(60) --- loglike-Train: -10574.65206537887 /// loglike-Test: -11898.020308972893 /// MAE: 116.53607132577864 /// Mean CRPS: 4012.5567255434785

seasonmodel_(2)_summer_states(5) --- loglike-Train: -9372.221248924981 /// loglike-Test: -14415.232962838378 /// MAE: 150.2796933454855 /// Mean CRPS: 5267.382925724638
seasonmodel_(2)_summer_states(10) --- loglike-Train: -8693.741977828406 /// loglike-Test: -12942.29249117585 /// MAE: 133.37016250633857 /// Mean CRPS: 4584.256567028986
seasonmodel_(2)_summer_states(15) --- loglike-Train: -8364.828193241616 /// loglike-Test: -12617.549175406719 /// MAE: 124.87880245133701 /// Mean CRPS: 4286.668251811594
seasonmodel_(2)_summer_states(20) --- loglike-Train: -8373.855154007393 /// loglike-Test: -12640.717755981557 /// MAE: 128.5835952248324 /// Mean CRPS: 4319.413496376812
seasonmodel_(2)_summer_states(25) --- loglike-Train: -8364.080708570995 /// loglike-Test: -12710.31429377606 /// MAE: 126.77976401228345 /// Mean CRPS: 4290.473845108696
seasonmodel_(2)_summer_states(30) --- loglike-Train: -8256.53532175453 /// loglike-Test: -12914.095763099785 /// MAE: 128.8238648955811 /// Mean CRPS: 4395.524003623188
seasonmodel_(2)_summer_states(40) --- loglike-Train: -8131.5447888322715 /// loglike-Test: -12987.657713636687 /// MAE: 125.75026080412931 /// Mean CRPS: 4316.05910326087
seasonmodel_(2)_summer_states(50) --- loglike-Train: -7966.222325239401 /// loglike-Test: -13373.140518383925 /// MAE: 129.6346301221585 /// Mean CRPS: 4451.47509059791
seasonmodel_(2)_summer_states(60) --- loglike-Train: -7955.43757651064 /// loglike-Test: -13375.927680873749 /// MAE: 128.08937090585866 /// Mean CRPS: 4435.738224637681

seasonmodel_(2)_fall_states(5) --- loglike-Train: -10648.601316565724 /// loglike-Test: -13914.956474554152 /// MAE: 148.11101436249214 /// Mean CRPS: 5219.960279304029
seasonmodel_(2)_fall_states(10) --- loglike-Train: -10070.629613654975 /// loglike-Test: -13359.418672358464 /// MAE: 138.8784165847735 /// Mean CRPS: 4670.2101648351645
seasonmodel_(2)_fall_states(15) --- loglike-Train: -9783.646365540586 /// loglike-Test: -13231.357445145557 /// MAE: 128.13649499000613 /// Mean CRPS: 4473.769917582417
seasonmodel_(2)_fall_states(20) --- loglike-Train: -9758.619906905275 /// loglike-Test: -13250.677329505721 /// MAE: 132.89320445505064 /// Mean CRPS: 4522.683264652014
seasonmodel_(2)_fall_states(25) --- loglike-Train: -9705.317870149836 /// loglike-Test: -13367.949224071639 /// MAE: 132.529496606096353 /// Mean CRPS: 4560.500114468865
seasonmodel_(2)_fall_states(30) --- loglike-Train: -9667.600058059168 /// loglike-Test: -13293.232481573825 /// MAE: 131.9883335908909 /// Mean CRPS: 4534.670673076923
seasonmodel_(2)_fall_states(40) --- loglike-Train: -9443.566021301402 /// loglike-Test: -13419.327486651124 /// MAE: 132.25575292599666 /// Mean CRPS: 4539.229166666667
seasonmodel_(2)_fall_states(50) --- loglike-Train: -9374.522132494687 /// loglike-Test: -13748.856309783656 /// MAE: 135.06250911882714 /// Mean CRPS: 4665.15293040293
seasonmodel_(2)_fall_states(60) --- loglike-Train: -9277.162013260124 /// loglike-Test: -14055.613324731414 /// MAE: 131.69507053412482 /// Mean CRPS: 4634.850961538462

seasonmodel_(2)_winter_states(5) --- loglike-Train: -11589.404718653 /// loglike-Test: -11081.821284440826 /// MAE: 122.31038397629239 /// Mean CRPS: 4031.7248168498168
seasonmodel_(2)_winter_states(10) --- loglike-Train: -10695.966670289545 /// loglike-Test: -10425.9981826736 /// MAE: 101.37454506586144 /// Mean CRPS: 3564.8341346153848
seasonmodel_(2)_winter_states(15) --- loglike-Train: -10613.051966925448 /// loglike-Test: -10401.055314719148 /// MAE: 100.76466833790968 /// Mean CRPS: 3497.665750915751
seasonmodel_(2)_winter_states(20) --- loglike-Train: -10600.417715483864 /// loglike-Test: -10446.606532738584 /// MAE: 102.31377681476962 /// Mean CRPS: 3590.4986978022
seasonmodel_(2)_winter_states(25) --- loglike-Train: -10358.252650884408 /// loglike-Test: -10585.34324647341 /// MAE: 98.86252239918424 /// Mean CRPS: 3482.137248168498
seasonmodel_(2)_winter_states(30) --- loglike-Train: -10388.180227027933 /// loglike-Test: -10583.299612903545 /// MAE: 102.42771189029136 /// Mean CRPS: 3494.0013736263736
seasonmodel_(2)_winter_states(40) --- loglike-Train: -10062.44089861155 /// loglike-Test: -10956.234269671662 /// MAE: 105.18590155749607 /// Mean CRPS: 3613.696085164835
seasonmodel_(2)_winter_states(50) --- loglike-Train: -9905.501380745107 /// loglike-Test: -11060.973328409946 /// MAE: 104.09341911861 /// Mean CRPS: 3577.8985805860807
seasonmodel_(2)_winter_states(60) --- loglike-Train: -9792.218519194117 /// loglike-Test: -11082.90098178272 /// MAE: 102.15744020067929 /// Mean CRPS: 3562.401785714286

----- Evaluate basismodel_hh(3) -----

basismodel_hh(3)_states(5) --- loglike-Train: -65173.97948889518 /// loglike-Test: -68025.22933722445 /// MAE: 321.49347996078876 /// Mean CRPS: 10546.688951502732
basismodel_hh(3)_states(10) --- loglike-Train: -63814.2849409459 /// loglike-Test: -66351.76241664586 /// MAE: 304.0381043900777 /// Mean CRPS: 10081.69840500091
basismodel_hh(3)_states(15) --- loglike-Train: -62784.13680623903 /// loglike-Test: -65645.02132411517 /// MAE: 288.75923381540605 /// Mean CRPS: 9713.45597108379
basismodel_hh(3)_states(20) --- loglike-Train: -62486.354821872585 /// loglike-Test: -65394.12426450156 /// MAE: 287.6943630798878 /// Mean CRPS: 9693.263689663023
basismodel_hh(3)_states(25) --- loglike-Train: -62260.438879135116 /// loglike-Test: -65394.646815994056 /// MAE: 287.26079090560665 /// Mean CRPS: 9681.75779826958
basismodel_hh(3)_states(30) --- loglike-Train: -62054.13157123463 /// loglike-Test: -65262.88543257005 /// MAE: 285.7339523553933 /// Mean CRPS: 9603.748491575592
basismodel_hh(3)_states(40) --- loglike-Train: -61155.86190563714 /// loglike-Test: -64885.35250489526 /// MAE: 284.4540999431862 /// Mean CRPS: 9557.663194444445
basismodel_hh(3)_states(50) --- loglike-Train: -61041.92723972328 /// loglike-Test: -65023.01333762864 /// MAE: 284.1766882753029 /// Mean CRPS: 9565.450620446267
basismodel_hh(3)_states(60) --- loglike-Train: -60689.65495212277 /// loglike-Test: -65138.71910617726 /// MAE: 284.2932007193083 /// Mean CRPS: 9608.024846311475

Timing: {1501.812}sec

----- Evaluate basismodel_hh(3) with timestamps -----

basismodel_ts(4)_states(5) --- loglike-Train: -92143.84094869546 /// loglike-Test: -96763.4108358905 /// MAE: 399.95758737814197 /// Mean CRPS: 13174.736879553733
basismodel_ts(4)_states(10) --- loglike-Train: -74725.62451062088 /// loglike-Test: -78890.25051540813 /// MAE: 367.19025043602903 /// Mean CRPS: 11818.180527094717
basismodel_ts(4)_states(15) --- loglike-Train: -71706.25894699615 /// loglike-Test: -75619.50499650778 /// MAE: 348.14569668919387 /// Mean CRPS: 11010.301627959927
basismodel_ts(4)_states(20) --- loglike-Train: -70457.45488393892 /// loglike-Test: -74677.56193436153 /// MAE: 332.7044618906249 /// Mean CRPS: 10577.901326275045
basismodel_ts(4)_states(25) --- loglike-Train: -69488.32699378712 /// loglike-Test: -73531.67199662357 /// MAE: 322.3095113400923 /// Mean CRPS: 10385.691171448087
basismodel_ts(4)_states(30) --- loglike-Train: -68755.75512016618 /// loglike-Test: -72974.95915699215 /// MAE: 309.1958583106432 /// Mean CRPS: 10064.67725409836
basismodel_ts(4)_states(40) --- loglike-Train: -67698.27645013496 /// loglike-Test: -71999.5983771896 /// MAE: 294.72585577688704 /// Mean CRPS: 9840.112363387978
basismodel_ts(4)_states(50) --- loglike-Train: -67638.16318720258 /// loglike-Test: -72136.72660922946 /// MAE: 298.5272382874903 /// Mean CRPS: 9888.515397313296
basismodel_ts(4)_states(60) --- loglike-Train: -68274.24691463762 /// loglike-Test: -73568.64661729685 /// MAE: 299.07409409216604 /// Mean CRPS: 9999.59539936248

basismodel_ts(8)_states(5) --- loglike-Train: -106761.6231069547 /// loglike-Test: -111494.4333614083 /// MAE: 392.4042872050527 /// Mean CRPS: 13123.283982240437
basismodel_ts(8)_states(10) --- loglike-Train: -86944.77750265544 /// loglike-Test: -91501.58735169588 /// MAE: 381.70879350571744 /// Mean CRPS: 12559.74447859745
basismodel_ts(8)_states(15) --- loglike-Train: -80928.4983247758 /// loglike-Test: -85525.74830159772 /// MAE: 365.2292076565263 /// Mean CRPS: 12020.221653005465
basismodel_ts(8)_states(20) --- loglike-Train: -79161.64692183107 /// loglike-Test: -83637.54234261953 /// MAE: 352.234626518466 /// Mean CRPS: 11500.872381602914
basismodel_ts(8)_states(25) --- loglike-Train: -76992.19836813859 /// loglike-Test: -81604.23669249103 /// MAE: 352.4846731868473 /// Mean CRPS: 11482.505151411657
basismodel_ts(8)_states(30) --- loglike-Train: -75892.17969320781 /// loglike-Test: -81183.52120500646 /// MAE: 340.7831659569395 /// Mean CRPS: 10993.779855418943
basismodel_ts(8)_states(40) --- loglike-Train: -73799.67661479078 /// loglike-Test: -84563.14524156286 /// MAE: 340.59144371568016 /// Mean CRPS: 11063.204461885246
basismodel_ts(8)_states(50) --- loglike-Train: -77244.475196156 /// loglike-Test: -84205.38109348572 /// MAE: 330.7204329836154 /// Mean CRPS: 10741.051229508197
basismodel_ts(8)_states(60) --- loglike-Train: -72812.07850366895 /// loglike-Test: -78858.04078109434 /// MAE: 320.0078517065182 /// Mean CRPS: 10369.037795992718

basismodel_ts(12)_states(5) --- loglike-Train: -122733.45127036796 /// loglike-Test: -127883.04385728129 /// MAE: 405.76780874861726 /// Mean CRPS: 13520.104849726777
basismodel_ts(12)_states(10) --- loglike-Train: -97132.65469389135 /// loglike-Test: -102018.16966257904 /// MAE: 385.0326214750006 /// Mean CRPS: 12966.33097108379
basismodel_ts(12)_states(15) --- loglike-Train: -89425.53531987063 /// loglike-Test: -94872.53315441322 /// MAE: 375.78699143835456 /// Mean CRPS: 12579.89412568306
basismodel_ts(12)_states(20) --- loglike-Train: -83827.17302628915 /// loglike-Test: -89151.61902034654 /// MAE: 362.1185545582546 /// Mean CRPS: 11819.357211976321
basismodel_ts(12)_states(25) --- loglike-Train: -82643.10405920868 /// loglike-Test: -87649.61613431422 /// MAE: 362.4578324209212 /// Mean CRPS: 11621.523110200364
basismodel_ts(12)_states(30) --- loglike-Train: -80755.07392043574 /// loglike-Test: -86364.86360107281 /// MAE: 349.3214516922382 /// Mean CRPS: 11256.0422928051
basismodel_ts(12)_states(40) --- loglike-Train: -79279.67661479078 /// loglike-Test: -84563.14524156286 /// MAE: 340.59144371568016 /// Mean CRPS: 11063.204461885246
basismodel_ts(12)_states(50) --- loglike-Train: -77244.475196156 /// loglike-Test: -84205.38109348572 /// MAE: 330.7204329836154 /// Mean CRPS: 10741.051229508197
basismodel_ts(12)_states(60) --- loglike-Train: -77212.5118172836 /// loglike-Test: -83761.53872321264 /// MAE: 332.4650236009767 /// Mean CRPS: 10848.65383082878

basismodel_ts(24)_states(5) --- loglike-Train: -152125.5945068324 /// loglike-Test: -159425.94453293222 /// MAE: 404.55714999303547 /// Mean CRPS: 13538.575791211293
basismodel_ts(24)_states(10) --- loglike-Train: -123296.8889855121 /// loglike-Test: -129846.17543157627 /// MAE: 391.73361837383993 /// Mean CRPS: 13128.457792577414
basismodel_ts(24)_states(15) --- loglike-Train: -112173.7398752776 /// loglike-Test: -118386.8327657342 /// MAE: 389.54341939019184 /// Mean CRPS: 12958.808430100182
basismodel_ts(24)_states(20) --- loglike-Train: -105122.27800011574 /// loglike-Test: -112701.2083370338 /// MAE: 380.74695346572815 /// Mean CRPS: 12750.975409836066
basismodel_ts(24)_states(25) --- loglike-Train: -98446.25608596412 /// loglike-Test: -105761.99245807475 /// MAE: 377.0545286877548 /// Mean CRPS: 12599.892873406194
basismodel_ts(24)_states(30) --- loglike-Train: -95193.09228405751 /// loglike-Test: -102975.68141517261 /// MAE: 374.6842711526414 /// Mean CRPS: 12457.167463570127
basismodel_ts(24)_states(40) --- loglike-Train: -90613.68718613937 /// loglike-Test: -100097.52232119776 /// MAE: 363.6876580955082 /// Mean CRPS: 12056.736281876138
basismodel_ts(24)_states(50) --- loglike-Train: -87783.22706778479 /// loglike-Test: -96548.02643199736 /// MAE: 359.8543710966926 /// Mean CRPS: 11747.077358018214
basismodel_ts(24)_states(60) --- loglike-Train: -86714.37422674744 /// loglike-Test: -96231.47607436948 /// MAE: 356.77438724129985 /// Mean CRPS: 11662.802339480875

basismodel_ts(96)_states(5) --- loglike-Train: -230111.94510172156 /// loglike-Test: -245972.97674426233 /// MAE: 407.75887925142143 /// Mean CRPS: 13688.30154116576
basismodel_ts(96)_states(10) --- loglike-Train: -230087.2751294701 /// loglike-Test: -245878.65725759562 /// MAE: 407.73583735631075 /// Mean CRPS: 13688.2284863066557
basismodel_ts(96)_states(15) --- loglike-Train: -230195.93246685815 /// loglike-Test: -245977.17017280863 /// MAE: 407.72873466796926 /// Mean CRPS: 13688.317964480875
basismodel_ts(96)_states(20) --- loglike-Train: -230628.89798693502 /// loglike-Test: -246302.08302208036 /// MAE: 407.7387272968631 /// Mean CRPS: 13688.478512067395
basismodel_ts(96)_states(25) --- loglike-Train: -230791.87589965342 /// loglike-Test: -246426.55640009029 /// MAE: 407.72901693520856 /// Mean CRPS: 13687.443249089254
basismodel_ts(96)_states(30) --- loglike-Train: -230906.00580679748 /// loglike-Test: -246527.54305490616 /// MAE: 407.71558391359326 /// Mean CRPS: 13688.048411885246
basismodel_ts(96)_states(40) --- loglike-Train: -230960.1487207171 /// loglike-Test: -246549.62000802928 /// MAE: 407.7284338038132 /// Mean CRPS: 13687.950648907105
basismodel_ts(96)_states(50) --- loglike-Train: -231087.65332669433 /// loglike-Test: -246676.66428201407 /// MAE: 407.7172497067348 /// Mean CRPS: 13688.140141165755
basismodel_ts(96)_states(60) --- loglike-Train: -231136.14163703407 /// loglike-Test: -246698.29877819063 /// MAE: 407.73812442735357 /// Mean CRPS: 13688.19489981785

Timing: {7963.32}sec

----- Evaluate seasonmodel_hh(3) -----

seasonmodel_(3)_spring_states(5) --- loglike-Train: -17110.24173506556 /// loglike-Test: -18120.663040031188 /// MAE: 347.8885471145081 /// Mean CRPS: 11521.763586956522
seasonmodel_(3)_spring_states(10) --- loglike-Train: -16523.03491155703 /// loglike-Test: -17621.358851665405 /// MAE: 331.9045572805671 /// Mean CRPS: 10986.186707427536
seasonmodel_(3)_spring_states(15) --- loglike-Train: -16254.856870819485 /// loglike-Test: -17511.063378049555 /// MAE: 314.6740392561849 /// Mean CRPS: 10711.595651594202
seasonmodel_(3)_spring_states(20) --- loglike-Train: -16115.205790732167 /// loglike-Test: -17603.545637327767 /// MAE: 317.46526177889956 /// Mean CRPS: 10797.1621318840508
seasonmodel_(3)_spring_states(25) --- loglike-Train: -16014.39622134164 /// loglike-Test: -17759.78997079864 /// MAE: 319.7832196449711 /// Mean CRPS: 10876.719089673914
seasonmodel_(3)_spring_states(30) --- loglike-Train: -15856.253453255411 /// loglike-Test: -17799.6403443484 /// MAE: 327.9509370083014 /// Mean CRPS: 11033.9617730072464
seasonmodel_(3)_spring_states(40) --- loglike-Train: -15571.488622319619 /// loglike-Test: -18066.62902054396 /// MAE: 320.9653950988714 /// Mean CRPS: 10929.35348731884
seasonmodel_(3)_spring_states(50) --- loglike-Train: -15422.8722700106 /// loglike-Test: -18226.227980856653 /// MAE: 331.38187047485326 /// Mean CRPS: 11257.732336956522
seasonmodel_(3)_spring_states(60) --- loglike-Train: -15240.374257029529 /// loglike-Test: -18536.564375298323 /// MAE: 326.22230033155535 /// Mean CRPS: 11190.74365942029

seasonmodel_(3)_summer_states(5) --- loglike-Train: -14445.242581305314 /// loglike-Test: -16502.152463999766 /// MAE: 300.48522348126664 /// Mean CRPS: 9724.831974637682
seasonmodel_(3)_summer_states(10) --- loglike-Train: -13692.902385550804 /// loglike-Test: -16161.992583122987 /// MAE: 279.2544726206632 /// Mean CRPS: 9255.962296195652
seasonmodel_(3)_summer_states(15) --- loglike-Train: -13465.258221105549 /// loglike-Test: -16335.754210517793 /// MAE: 286.6398598037038 /// Mean CRPS: 9354.366055253624
seasonmodel_(3)_summer_states(20) --- loglike-Train: -13298.84579793139 /// loglike-Test: -16452.106375498253 /// MAE: 284.69216207386785 /// Mean CRPS: 9368.773324275362
seasonmodel_(3)_summer_states(25) --- loglike-Train: -13179.652925771348 /// loglike-Test: -16196.476546796892 /// MAE: 273.0904402609785 /// Mean CRPS: 9092.236413043478
seasonmodel_(3)_summer_states(30) --- loglike-Train: -12970.103231513793 /// loglike-Test: -16464.434645592955 /// MAE: 278.5326092695695 /// Mean CRPS: 9289.588088768116
seasonmodel_(3)_summer_states(40) --- loglike-Train: -12765.519796777282 /// loglike-Test: -16656.089362589042 /// MAE: 285.14736895281214 /// Mean CRPS: 9366.037703804348
seasonmodel_(3)_summer_states(50) --- loglike-Train: -12662.982428955107 /// loglike-Test: -16906.268534987474 /// MAE: 284.8032800451394 /// Mean CRPS: 9434.61277173913
seasonmodel_(3)_summer_states(60) --- loglike-Train: -12538.51960774553 /// loglike-Test: -17017.099625436174 /// MAE: 286.661806161282 /// Mean CRPS: 9548.708333333334

seasonmodel_(3)_fall_states(5) --- loglike-Train: -16487.320993278638 /// loglike-Test: -17023.906799768025 /// MAE: 323.8775486337624 /// Mean CRPS: 10541.937843406593
seasonmodel_(3)_fall_states(10) --- loglike-Train: -16129.247209757064 /// loglike-Test: -16780.205390589657 /// MAE: 304.5997642524962 /// Mean CRPS: 10122.382211538461
seasonmodel_(3)_fall_states(15) --- loglike-Train: -15828.216065932456 /// loglike-Test: -16627.487878399214 /// MAE: 291.77817683158185 /// Mean CRPS: 9906.038003663003
seasonmodel_(3)_fall_states(20) --- loglike-Train: -15724.031575515895 /// loglike-Test: -16738.462933458155 /// MAE: 294.3057249789897 /// Mean CRPS: 9965.62248168498
seasonmodel_(3)_fall_states(25) --- loglike-Train: -15665.28029031478 /// loglike-Test: -16740.482129742843 /// MAE: 292.7697640563287 /// Mean CRPS: 9962.76076007326
seasonmodel_(3)_fall_states(30) --- loglike-Train: -15409.18633934382 /// loglike-Test: -16893.063577014313 /// MAE: 294.5198505539127 /// Mean CRPS: 10027.23076923077
seasonmodel_(3)_fall_states(40) --- loglike-Train: -15224.464936208304 /// loglike-Test: -17029.99878651651 /// MAE: 296.96843188519614 /// Mean CRPS: 10113.018658424908
seasonmodel_(3)_fall_states(50) --- loglike-Train: -15017.562979001395 /// loglike-Test: -17108.085492212962 /// MAE: 297.7987480611891 /// Mean CRPS: 10136.080586080587
seasonmodel_(3)_fall_states(60) --- loglike-Train: -14812.64813141046 /// loglike-Test: -17395.05958461251 /// MAE: 302.079549109591 /// Mean CRPS: 10329.586538461539

seasonmodel_(3)_winter_states(5) --- loglike-Train: -16875.915324287795 /// loglike-Test: -17008.69709781965 /// MAE: 323.09459880537696 /// Mean CRPS: 10552.136675824177
seasonmodel_(3)_winter_states(10) --- loglike-Train: -16379.394842391466 /// loglike-Test: -16675.761777448606 /// MAE: 311.01882193952 /// Mean CRPS: 10182.486836080587
seasonmodel_(3)_winter_states(15) --- loglike-Train: -16080.251527295157 /// loglike-Test: -16634.913631620853 /// MAE: 292.6879801790539 /// Mean CRPS: 9870.714858058607
seasonmodel_(3)_winter_states(20) --- loglike-Train: -15901.004119455256 /// loglike-Test: -16653.79747656143 /// MAE: 294.03678624466437 /// Mean CRPS: 9866.68646978022
seasonmodel_(3)_winter_states(25) --- loglike-Train: -15798.49760544919 /// loglike-Test: -16827.42901194311 /// MAE: 299.73561266189995 /// Mean CRPS: 9956.551167582418
seasonmodel_(3)_winter_states(30) --- loglike-Train: -15649.056356919451 /// loglike-Test: -16988.989177546806 /// MAE: 300.55859055401777 /// Mean CRPS: 10023.03032429084
seasonmodel_(3)_winter_states(40) --- loglike-Train: -15402.628623127472 /// loglike-Test: -17365.49001236978 /// MAE: 303.32127758390953 /// Mean CRPS: 10162.653502747253
seasonmodel_(3)_winter_states(50) --- loglike-Train: -15179.8700720961 /// loglike-Test: -17298.803314008095 /// MAE: 299.2366247194452 /// Mean CRPS: 10174.283195970696
seasonmodel_(3)_winter_states(60) --- loglike-Train: -15046.389411760894 /// loglike-Test: -17576.29259893637 /// MAE: 309.3218442208479 /// Mean CRPS: 10356.467605311354

Timing: {350.339}sec

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----- Evaluate basismodel_hh(4) -----
basismodel_hh(4)_states(5) --- loglike-Train: -51975.59021227203 /// loglike-Test: -59853.809396994395 /// MAE: 256.0960772937634 /// Mean CRPS: 8023.674863387978
basismodel_hh(4)_states(10) --- loglike-Train: -49514.537890232124 /// loglike-Test: -58093.30301545003 /// MAE: 215.8725652101133 /// Mean CRPS: 7225.469233834244
basismodel_hh(4)_states(15) --- loglike-Train: -49604.15226383946 /// loglike-Test: -57414.13160956888 /// MAE: 213.77976157289336 /// Mean CRPS: 7087.218806921675
basismodel_hh(4)_states(20) --- loglike-Train: -48497.765425323436 /// loglike-Test: -55365.60910272534 /// MAE: 221.97962903689796 /// Mean CRPS: 7081.594575364299
basismodel_hh(4)_states(25) --- loglike-Train: -48097.19776823378 /// loglike-Test: -54762.71057050233 /// MAE: 202.63597160484275 /// Mean CRPS: 6823.998121584699
basismodel_hh(4)_states(30) --- loglike-Train: -48126.5838244436 /// loglike-Test: -54629.93922382353 /// MAE: 206.96159914659466 /// Mean CRPS: 6862.2271714786339
basismodel_hh(4)_states(35) --- loglike-Train: -47715.621144011704 /// loglike-Test: -54141.06872850904 /// MAE: 207.86570312161754 /// Mean CRPS: 6808.728512067395
basismodel_hh(4)_states(40) --- loglike-Train: -47968.480751576775 /// loglike-Test: -55917.7152857523 /// MAE: 217.7257608188153 /// Mean CRPS: 7054.405766165756
basismodel_hh(4)_states(60) --- loglike-Train: -47347.07642266042 /// loglike-Test: -54332.7931923351 /// MAE: 205.6238394634366 /// Mean CRPS: 6849.707109517304
### Timing: [1627.745]sec ###
----- Evaluate basismodel_hh(4) with timestamps -----
basismodel_ts(4)_states(5) --- loglike-Train: -73465.58785865639 /// loglike-Test: -84842.85422517572 /// MAE: 273.00534894951943 /// Mean CRPS: 9642.752362249545
basismodel_ts(4)_states(10) --- loglike-Train: -60716.83940359736 /// loglike-Test: -70932.89847879714 /// MAE: 259.2661222689362 /// Mean CRPS: 8888.452688752277
basismodel_ts(4)_states(20) --- loglike-Train: -55314.53389800455 /// loglike-Test: -65497.38231493413 /// MAE: 234.1122055028903 /// Mean CRPS: 7493.666609744991
basismodel_ts(4)_states(30) --- loglike-Train: -57290.67794104717 /// loglike-Test: -67992.70869296916 /// MAE: 253.19890386630868 /// Mean CRPS: 8282.111594945356
basismodel_ts(4)_states(40) --- loglike-Train: -54602.32795732454 /// loglike-Test: -64927.5046933621 /// MAE: 224.48294543234846 /// Mean CRPS: 7347.27065064663
basismodel_ts(4)_states(50) --- loglike-Train: -54196.92055400452 /// loglike-Test: -64083.3265979191 /// MAE: 216.50589180952693 /// Mean CRPS: 7174.1092326958105
basismodel_ts(4)_states(60) --- loglike-Train: -53575.157131857384 /// loglike-Test: -64088.65651875413 /// MAE: 224.9975820748446 /// Mean CRPS: 7257.565175318761
basismodel_ts(4)_states(70) --- loglike-Train: -53196.401631587345 /// loglike-Test: -65279.42613241093 /// MAE: 221.0925697608898 /// Mean CRPS: 7234.9955010929
basismodel_ts(4)_states(80) --- loglike-Train: -52527.024562226 /// loglike-Test: -64132.041022530604 /// MAE: 219.5499705971081 /// Mean CRPS: 7195.662454462659
-----
basismodel_ts(8)_states(5) --- loglike-Train: -89540.88714379471 /// loglike-Test: -102624.1191959819 /// MAE: 258.31180929799604 /// Mean CRPS: 9584.225637522768
basismodel_ts(8)_states(10) --- loglike-Train: -71616.13095762012 /// loglike-Test: -84404.29133978872 /// MAE: 267.9767067303141 /// Mean CRPS: 9350.338740892532
basismodel_ts(8)_states(15) --- loglike-Train: -64646.70971717297 /// loglike-Test: -78560.59001639951 /// MAE: 266.3114598223035 /// Mean CRPS: 9354.470002276867
basismodel_ts(8)_states(20) --- loglike-Train: -61721.96153093602 /// loglike-Test: -75066.32539834818 /// MAE: 245.9480544022016 /// Mean CRPS: 8291.881631375227
basismodel_ts(8)_states(25) --- loglike-Train: -60478.69640999559 /// loglike-Test: -72839.62209882546 /// MAE: 251.25050496619858 /// Mean CRPS: 8072.573115892532
basismodel_ts(8)_states(30) --- loglike-Train: -59851.13300646384 /// loglike-Test: -73511.34508970693 /// MAE: 247.2460791258368 /// Mean CRPS: 7961.107318397856
basismodel_ts(8)_states(40) --- loglike-Train: -58297.974532004475 /// loglike-Test: -71850.01750392678 /// MAE: 245.79780485885757 /// Mean CRPS: 7796.834101775957
basismodel_ts(8)_states(50) --- loglike-Train: -57576.60182585889 /// loglike-Test: -71982.4362842305 /// MAE: 232.83581603838604 /// Mean CRPS: 7556.074738160291
basismodel_ts(8)_states(60) --- loglike-Train: -57798.15044438322 /// loglike-Test: -72876.4968481432 /// MAE: 232.80224660293683 /// Mean CRPS: 7593.860627276867
-----
basismodel_ts(12)_states(5) --- loglike-Train: -99226.46997910092 /// loglike-Test: -112485.83672848884 /// MAE: 264.9583312846334 /// Mean CRPS: 9634.428933287796
basismodel_ts(12)_states(10) --- loglike-Train: -108116.79188482794 /// loglike-Test: -126867.8158028259 /// MAE: 262.36044052917595 /// Mean CRPS: 9549.65229394534
basismodel_ts(12)_states(15) --- loglike-Train: -73290.65811508682 /// loglike-Test: -87177.20794036344 /// MAE: 264.2205332148559 /// Mean CRPS: 9361.842583105647
basismodel_ts(12)_states(20) --- loglike-Train: -67339.83500204331 /// loglike-Test: -81086.62751018535 /// MAE: 258.22870988045906 /// Mean CRPS: 8607.352857468124
basismodel_ts(12)_states(25) --- loglike-Train: -69037.13644460571 /// loglike-Test: -84972.58991621653 /// MAE: 264.41261059958146 /// Mean CRPS: 8872.506887522768
basismodel_ts(12)_states(30) --- loglike-Train: -64328.77108399605 /// loglike-Test: -79292.52869490927 /// MAE: 263.3643704820202 /// Mean CRPS: 8674.77423132969
basismodel_ts(12)_states(40) --- loglike-Train: -63156.39063896915 /// loglike-Test: -79798.75554506651 /// MAE: 252.83069718756087 /// Mean CRPS: 8497.838769353337
basismodel_ts(12)_states(50) --- loglike-Train: -61363.04837981835 /// loglike-Test: -77155.00188220228 /// MAE: 248.6231029023456 /// Mean CRPS: 7932.9644239526415
basismodel_ts(12)_states(60) --- loglike-Train: -60754.93545015489 /// loglike-Test: -77774.38147694242 /// MAE: 239.86659660384487 /// Mean CRPS: 7801.122324681239
-----
basismodel_ts(24)_states(5) --- loglike-Train: -133885.54195947078 /// loglike-Test: -152823.29015438075 /// MAE: 275.59158395278195 /// Mean CRPS: 10180.071322885945
basismodel_ts(24)_states(10) --- loglike-Train: -108116.79188482794 /// loglike-Test: -126867.8158028259 /// MAE: 262.36044052917595 /// Mean CRPS: 9543.401183970856
basismodel_ts(24)_states(15) --- loglike-Train: -98911.06108310763 /// loglike-Test: -118451.63651025857 /// MAE: 265.08911709503826 /// Mean CRPS: 9688.80268101093
basismodel_ts(24)_states(20) --- loglike-Train: -88081.004586117904 /// loglike-Test: -108252.30208371075 /// MAE: 263.2206285633237 /// Mean CRPS: 9529.178107923497
basismodel_ts(24)_states(25) --- loglike-Train: -83457.3761625688 /// loglike-Test: -103529.56517190422 /// MAE: 261.57638562927565 /// Mean CRPS: 9315.581938752277
basismodel_ts(24)_states(30) --- loglike-Train: -76958.36768740896 /// loglike-Test: -98210.10667487049 /// MAE: 263.6099732606352 /// Mean CRPS: 9224.4356215847
basismodel_ts(24)_states(40) --- loglike-Train: -73898.5126557931 /// loglike-Test: -94210.6857199993 /// MAE: 264.82274045628066 /// Mean CRPS: 9222.608321948997
basismodel_ts(24)_states(50) --- loglike-Train: -70593.82941222028 /// loglike-Test: -92495.98050856421 /// MAE: 259.032387366944 /// Mean CRPS: 8756.828722677596
basismodel_ts(24)_states(60) --- loglike-Train: -68402.2595717975 /// loglike-Test: -91089.98368082457 /// MAE: 258.0106256223428 /// Mean CRPS: 8630.571522085611
-----
basismodel_ts(96)_states(5) --- loglike-Train: -174194.02570468173 /// loglike-Test: -208738.74904070288 /// MAE: 275.8426687379407 /// Mean CRPS: 10185.795081967213
basismodel_ts(96)_states(10) --- loglike-Train: -156990.80121033272 /// loglike-Test: -192139.9498338109 /// MAE: 273.3157425693734 /// Mean CRPS: 10101.81210154827
basismodel_ts(96)_states(15) --- loglike-Train: -146277.90080345792 /// loglike-Test: -184222.73006603774 /// MAE: 274.5277829259645 /// Mean CRPS: 10143.152436247723
basismodel_ts(96)_states(20) --- loglike-Train: -134374.16457824013 /// loglike-Test: -170893.62487172266 /// MAE: 275.2557101957506 /// Mean CRPS: 10163.926627959927
basismodel_ts(96)_states(25) --- loglike-Train: -123383.5383622292 /// loglike-Test: -159044.2562408233 /// MAE: 272.7257350501447 /// Mean CRPS: 10023.633794398907
basismodel_ts(96)_states(30) --- loglike-Train: -121758.74996473102 /// loglike-Test: -159358.16915601242 /// MAE: 273.7322436515844 /// Mean CRPS: 10034.623833105647
basismodel_ts(96)_states(40) --- loglike-Train: -102767.1583752766 /// loglike-Test: -138706.56465376608 /// MAE: 270.28352929270767 /// Mean CRPS: 9977.876166894353
basismodel_ts(96)_states(50) --- loglike-Train: -96572.84332488784 /// loglike-Test: -133555.56773972092 /// MAE: 272.3708139651541 /// Mean CRPS: 9998.139571948997
basismodel_ts(96)_states(60) --- loglike-Train: -87941.6458436714 /// loglike-Test: -123934.2966645089 /// MAE: 272.0493745821963 /// Mean CRPS: 9989.495617030965
-----
### Timing: [8406.558]sec ###
----- Evaluate seasonmodel_hh(4) -----
seasonmodel_(4)_spring_states(5) --- loglike-Train: -12595.268465654448 /// loglike-Test: -15482.928993265941 /// MAE: 252.2707607942099 /// Mean CRPS: 8306.65036231884
seasonmodel_(4)_spring_states(10) --- loglike-Train: -11746.418260160692 /// loglike-Test: -14959.89834647917 /// MAE: 234.70273896885783 /// Mean CRPS: 7861.147078804348
seasonmodel_(4)_spring_states(15) --- loglike-Train: -11792.224918546093 /// loglike-Test: -15029.018434899208 /// MAE: 228.74742848272163 /// Mean CRPS: 7601.61231884058
seasonmodel_(4)_spring_states(20) --- loglike-Train: -11292.601947868232 /// loglike-Test: -14880.49146350374 /// MAE: 223.62878585973763 /// Mean CRPS: 7689.538949275362
seasonmodel_(4)_spring_states(25) --- loglike-Train: -11133.901997124354 /// loglike-Test: -15112.634324921373 /// MAE: 231.59982749304228 /// Mean CRPS: 7639.156589673913
seasonmodel_(4)_spring_states(30) --- loglike-Train: -11045.346127946326 /// loglike-Test: -14998.666704997853 /// MAE: 224.57734555495156 /// Mean CRPS: 7763.316689311594
seasonmodel_(4)_spring_states(40) --- loglike-Train: -11022.849440391597 /// loglike-Test: -15346.327983269137 /// MAE: 233.18670372129145 /// Mean CRPS: 7920.234488226348
seasonmodel_(4)_spring_states(50) --- loglike-Train: -10848.699809407779 /// loglike-Test: -15544.978308262185 /// MAE: 239.3659470971309 /// Mean CRPS: 8044.245131340558
seasonmodel_(4)_spring_states(60) --- loglike-Train: -10608.244181396596 /// loglike-Test: -16072.289997513213 /// MAE: 240.43073114326536 /// Mean CRPS: 8206.622169384058
-----
seasonmodel_(4)_summer_states(5) --- loglike-Train: -12815.258363545154 /// loglike-Test: -17391.690092697798 /// MAE: 273.36845154682726 /// Mean CRPS: 8213.631227355072
seasonmodel_(4)_summer_states(10) --- loglike-Train: -12449.743957742765 /// loglike-Test: -17189.413538008364 /// MAE: 213.23718157341057 /// Mean CRPS: 7561.091259057971
seasonmodel_(4)_summer_states(15) --- loglike-Train: -12326.252099103629 /// loglike-Test: -17038.85184636187 /// MAE: 217.89261189348684 /// Mean CRPS: 7582.858922101449
seasonmodel_(4)_summer_states(20) --- loglike-Train: -12251.33912381099 /// loglike-Test: -17375.096551487295 /// MAE: 214.60536056310687 /// Mean CRPS: 7625.718976449275
seasonmodel_(4)_summer_states(25) --- loglike-Train: -12122.734481807485 /// loglike-Test: -17231.812260929575 /// MAE: 220.2573174064109 /// Mean CRPS: 7886.354279891304
seasonmodel_(4)_summer_states(30) --- loglike-Train: -12052.880585408244 /// loglike-Test: -17771.26505951043 /// MAE: 207.11390664675713 /// Mean CRPS: 7679.189990940209
seasonmodel_(4)_summer_states(40) --- loglike-Train: -11904.18877565099 /// loglike-Test: -17435.352473447463 /// MAE: 215.90231101910595 /// Mean CRPS: 7827.416440217391
seasonmodel_(4)_summer_states(50) --- loglike-Train: -11825.125963297409 /// loglike-Test: -17900.649072325115 /// MAE: 242.47605510913084 /// Mean CRPS: 8043.857568864579
seasonmodel_(4)_summer_states(60) --- loglike-Train: -11552.251842953978 /// loglike-Test: -16503.456711298903 /// MAE: 268.5948968041145 /// Mean CRPS: 8057.335144927536
-----
seasonmodel_(4)_fall_states(5) --- loglike-Train: -13020.735420838822 /// loglike-Test: -14603.893557962767 /// MAE: 244.9574922678516 /// Mean CRPS: 7810.83298992674
seasonmodel_(4)_fall_states(10) --- loglike-Train: -12098.703287526332 /// loglike-Test: -13850.534969300901 /// MAE: 214.97058539258018 /// Mean CRPS: 7060.65750915751
seasonmodel_(4)_fall_states(15) --- loglike-Train: -11936.015349059755 /// loglike-Test: -13731.65705142374 /// MAE: 213.45943409525853 /// Mean CRPS: 7034.943452380952
seasonmodel_(4)_fall_states(20) --- loglike-Train: -11854.670941427852 /// loglike-Test: -13584.171434837912 /// MAE: 216.00783232585079 /// Mean CRPS: 7033.345924908425
seasonmodel_(4)_fall_states(25) --- loglike-Train: -11635.367715859204 /// loglike-Test: -13836.143015037149 /// MAE: 217.93721482700946 /// Mean CRPS: 7073.684409340659
seasonmodel_(4)_fall_states(30) --- loglike-Train: -11708.003745799877 /// loglike-Test: -13802.614681044579 /// MAE: 217.47279289554575 /// Mean CRPS: 7142.39858058608
seasonmodel_(4)_fall_states(40) --- loglike-Train: -11484.830923415346 /// loglike-Test: -13788.535759868446 /// MAE: 215.7125055634214 /// Mean CRPS: 7034.054716117216
seasonmodel_(4)_fall_states(50) --- loglike-Train: -11454.168441653484 /// loglike-Test: -13835.195549300475 /// MAE: 216.49321420583337 /// Mean CRPS: 7086.73385989011
seasonmodel_(4)_fall_states(60) --- loglike-Train: -11042.272985953288 /// loglike-Test: -14340.491327452874 /// MAE: 215.7345927924973 /// Mean CRPS: 7243.832646520146
-----
seasonmodel_(4)_winter_states(5) --- loglike-Train: -12792.111117244665 /// loglike-Test: -13325.674806958268 /// MAE: 233.6709517884712 /// Mean CRPS: 7679.447916666667
seasonmodel_(4)_winter_states(10) --- loglike-Train: -12135.811148869665 /// loglike-Test: -12842.701047799233 /// MAE: 222.68855046421012 /// Mean CRPS: 7261.783653846154
seasonmodel_(4)_winter_states(15) --- loglike-Train: -11946.999550325745 /// loglike-Test: -12690.816437819249 /// MAE: 206.51129793074801 /// Mean CRPS: 6846.191506410257
seasonmodel_(4)_winter_states(20) --- loglike-Train: -11873.165671025565 /// loglike-Test: -12719.1483061223 /// MAE: 210.66120814747936 /// Mean CRPS: 6968.5327380952385
seasonmodel_(4)_winter_states(25) --- loglike-Train: -11725.563041026111 /// loglike-Test: -12846.06913017229 /// MAE: 209.05475392324195 /// Mean CRPS: 6962.382549455055
seasonmodel_(4)_winter_states(30) --- loglike-Train: -11653.171941009103 /// loglike-Test: -12870.707734712625 /// MAE: 210.3677250871729 /// Mean CRPS: 6976.726991758242
seasonmodel_(4)_winter_states(40) --- loglike-Train: -11561.129044361163 /// loglike-Test: -12866.082754849085 /// MAE: 209.487285674352 /// Mean CRPS: 6971.251144688645
seasonmodel_(4)_winter_states(50) --- loglike-Train: -11490.295821327765 /// loglike-Test: -13005.438606960099 /// MAE: 213.42421210463381 /// Mean CRPS: 7047.826923076923
seasonmodel_(4)_winter_states(60) --- loglike-Train: -11158.230949446406 /// loglike-Test: -13549.749320207784 /// MAE: 209.12706572254876 /// Mean CRPS: 7127.172847985348
-----
### Timing: [362.528]sec ###
```

----- Evaluate basismodel_hh(5) -----

basismodel_hh(5)_states(5) --- loglike-Train: -40830.99182871838 /// loglike-Test: -41320.38076789221 /// MAE: 106.6756757237731 /// Mean CRPS: 3711.0873178506376
basismodel_hh(5)_states(10) --- loglike-Train: -35682.59737096286 /// loglike-Test: -37722.60527893657 /// MAE: 99.93223758950778 /// Mean CRPS: 3337.587659380692
basismodel_hh(5)_states(15) --- loglike-Train: -35467.91422443872 /// loglike-Test: -36810.53772685182 /// MAE: 97.70003891256242 /// Mean CRPS: 3264.9292748178505
basismodel_hh(5)_states(20) --- loglike-Train: -34132.11142511106 /// loglike-Test: -36439.0103915221 /// MAE: 97.69013857144813 /// Mean CRPS: 3271.9436190801457
basismodel_hh(5)_states(25) --- loglike-Train: -33520.97281845905 /// loglike-Test: -35614.15051090222 /// MAE: 94.84533778436641 /// Mean CRPS: 3168.5646630236793
basismodel_hh(5)_states(30) --- loglike-Train: -33389.797738923466 /// loglike-Test: -36027.0037019375 /// MAE: 94.39866891984636 /// Mean CRPS: 3157.9487704918033
basismodel_hh(5)_states(40) --- loglike-Train: -32850.44081490257 /// loglike-Test: -35351.48431478902 /// MAE: 94.98508354164728 /// Mean CRPS: 3171.81102003643
basismodel_hh(5)_states(50) --- loglike-Train: -32578.055678233643 /// loglike-Test: -35507.121917365934 /// MAE: 95.3658990833509 /// Mean CRPS: 3176.4184596994537
basismodel_hh(5)_states(60) --- loglike-Train: -32261.988278807865 /// loglike-Test: -35334.91383485356 /// MAE: 92.99526642844849 /// Mean CRPS: 3119.1683173952642
Timing: {1638.55}sec

----- Evaluate basismodel_hh(5) with timestamps -----

basismodel_ts(4)_states(5) --- loglike-Train: -78348.59591322628 /// loglike-Test: -75004.84036679743 /// MAE: 180.04333667792716 /// Mean CRPS: 5722.265340391621
basismodel_ts(4)_states(10) --- loglike-Train: -51152.155794589635 /// loglike-Test: -51128.871980111886 /// MAE: 129.75651631680307 /// Mean CRPS: 4095.410519125683
basismodel_ts(4)_states(15) --- loglike-Train: -50407.26034611958 /// loglike-Test: -51643.005773266486 /// MAE: 127.12919090025014 /// Mean CRPS: 4272.293146630237
basismodel_ts(4)_states(20) --- loglike-Train: -44082.41914804242 /// loglike-Test: -45615.23766345433 /// MAE: 103.03501859513258 /// Mean CRPS: 3499.245702413479
basismodel_ts(4)_states(25) --- loglike-Train: -42745.06264818538 /// loglike-Test: -44845.48347808249 /// MAE: 103.66298205871766 /// Mean CRPS: 3491.9822404371585
basismodel_ts(4)_states(30) --- loglike-Train: -41858.97103052627 /// loglike-Test: -43698.54159482713 /// MAE: 96.82204358686388 /// Mean CRPS: 3292.162027550091
basismodel_ts(4)_states(40) --- loglike-Train: -40716.64750619515 /// loglike-Test: -43626.98695486267 /// MAE: 98.90014030827516 /// Mean CRPS: 3313.219404592715
basismodel_ts(4)_states(50) --- loglike-Train: -39179.693122571436 /// loglike-Test: -42556.792478937554 /// MAE: 97.98524720989327 /// Mean CRPS: 3290.3051001821495
basismodel_ts(4)_states(60) --- loglike-Train: -38864.29129124597 /// loglike-Test: -42626.450893740366 /// MAE: 97.54205882383151 /// Mean CRPS: 3253.1608321948997

basismodel_ts(8)_states(5) --- loglike-Train: -90118.63918672349 /// loglike-Test: -86586.50456607915 /// MAE: 176.33342849548035 /// Mean CRPS: 5647.85695582878
basismodel_ts(8)_states(10) --- loglike-Train: -71998.99399939232 /// loglike-Test: -70373.4468948551 /// MAE: 169.73890283610555 /// Mean CRPS: 5433.123036202186
basismodel_ts(8)_states(15) --- loglike-Train: -63160.80160003794 /// loglike-Test: -62457.598981966046 /// MAE: 129.061927723522 /// Mean CRPS: 4433.035120673952
basismodel_ts(8)_states(20) --- loglike-Train: -57590.48851621695 /// loglike-Test: -57939.532416185015 /// MAE: 127.18952283428773 /// Mean CRPS: 4260.780936930783
basismodel_ts(8)_states(25) --- loglike-Train: -52296.69334915607 /// loglike-Test: -53595.97145742667 /// MAE: 116.6919947452646 /// Mean CRPS: 3825.7333219489983
basismodel_ts(8)_states(30) --- loglike-Train: -52705.83118591366 /// loglike-Test: -54306.288020552834 /// MAE: 128.50311420548996 /// Mean CRPS: 4069.3905396174864
basismodel_ts(8)_states(40) --- loglike-Train: -47568.61815455422 /// loglike-Test: -49570.14027104643 /// MAE: 101.07543718618669 /// Mean CRPS: 3404.53313694731385
basismodel_ts(8)_states(50) --- loglike-Train: -45764.664205912595 /// loglike-Test: -48447.81719962687 /// MAE: 102.17842513428118 /// Mean CRPS: 3468.7984972677596
basismodel_ts(8)_states(60) --- loglike-Train: -44794.49627021185 /// loglike-Test: -48349.925141057465 /// MAE: 97.57435091680057 /// Mean CRPS: 3231.6019182604737
basismodel_ts(12)_states(40) --- loglike-Train: -55655.226435451856 /// loglike-Test: -56934.33649606374 /// MAE: 113.87423553595504 /// Mean CRPS: 3686.8879496812388
basismodel_ts(12)_states(50) --- loglike-Train: -51955.86183524122 /// loglike-Test: -54399.31282586086 /// MAE: 105.0837851670722 /// Mean CRPS: 3548.064321493625
basismodel_ts(12)_states(60) --- loglike-Train: -51349.345881767294 /// loglike-Test: -54793.94974828895 /// MAE: 101.99380162691664 /// Mean CRPS: 3470.632058287796

basismodel_ts(12)_states(5) --- loglike-Train: -113082.74906724246 /// loglike-Test: -110527.02032475473 /// MAE: 177.12035960255926 /// Mean CRPS: 5473.273451730419
basismodel_ts(12)_states(10) --- loglike-Train: -89133.3959728917 /// loglike-Test: -86802.5306821726 /// MAE: 158.95627047808902 /// Mean CRPS: 5116.3832251821495
basismodel_ts(12)_states(15) --- loglike-Train: -81259.41815258452 /// loglike-Test: -79321.0648661395 /// MAE: 170.95409523258186 /// Mean CRPS: 5376.2026217049185
basismodel_ts(12)_states(20) --- loglike-Train: -69794.22496571895 /// loglike-Test: -69334.88664240466 /// MAE: 145.15842265569833 /// Mean CRPS: 4616.87476092896
basismodel_ts(12)_states(25) --- loglike-Train: -62523.26535997884 /// loglike-Test: -63126.92941518633 /// MAE: 132.3054612736236 /// Mean CRPS: 4306.986765710382
basismodel_ts(12)_states(30) --- loglike-Train: -59891.54657641282 /// loglike-Test: -59884.99563662111 /// MAE: 129.18171511448816 /// Mean CRPS: 4099.8272427140255
basismodel_ts(12)_states(40) --- loglike-Train: -55655.226435451856 /// loglike-Test: -56934.33649606374 /// MAE: 113.87423553595504 /// Mean CRPS: 3686.8879496812388
basismodel_ts(12)_states(50) --- loglike-Train: -51955.86183524122 /// loglike-Test: -54399.31282586086 /// MAE: 105.0837851670722 /// Mean CRPS: 3548.064321493625
basismodel_ts(12)_states(60) --- loglike-Train: -51349.345881767294 /// loglike-Test: -54793.94974828895 /// MAE: 101.99380162691664 /// Mean CRPS: 3470.632058287796

basismodel_ts(24)_states(5) --- loglike-Train: -140162.2702713942 /// loglike-Test: -140118.56545734912 /// MAE: 189.02347856321072 /// Mean CRPS: 6147.1543431238615
basismodel_ts(24)_states(10) --- loglike-Train: -114581.34874941534 /// loglike-Test: -114135.092126444503 /// MAE: 166.28614717406892 /// Mean CRPS: 5308.524561703097
basismodel_ts(24)_states(15) --- loglike-Train: -104723.65561937593 /// loglike-Test: -105614.64768782453 /// MAE: 179.37117947398986 /// Mean CRPS: 5651.77464708561
basismodel_ts(24)_states(20) --- loglike-Train: -90201.99962281699 /// loglike-Test: -97661.08371757055 /// MAE: 174.86230430096325 /// Mean CRPS: 5477.561361566484
basismodel_ts(24)_states(25) --- loglike-Train: -89003.33400243426 /// loglike-Test: -91478.29119628543 /// MAE: 154.82768346587562 /// Mean CRPS: 4992.8089139344265
basismodel_ts(24)_states(30) --- loglike-Train: -84283.39872768494 /// loglike-Test: -85209.9206332754 /// MAE: 142.12762115802127 /// Mean CRPS: 4789.009676684882
basismodel_ts(24)_states(40) --- loglike-Train: -74191.76957131324 /// loglike-Test: -75657.75960686491 /// MAE: 140.6126263782024 /// Mean CRPS: 4344.78890596392
basismodel_ts(24)_states(50) --- loglike-Train: -71873.01186659437 /// loglike-Test: -76085.81742009736 /// MAE: 123.6500901539537 /// Mean CRPS: 4036.1421903460837
basismodel_ts(24)_states(60) --- loglike-Train: -65678.51536626258 /// loglike-Test: -70289.81232353656 /// MAE: 117.7115566060228 /// Mean CRPS: 3857.9150159380692

basismodel_ts(96)_states(5) --- loglike-Train: -184435.6639079764 /// loglike-Test: -192128.53540858487 /// MAE: 201.8680143046042 /// Mean CRPS: 6531.995787795992
basismodel_ts(96)_states(10) --- loglike-Train: -166778.90549380326 /// loglike-Test: -176624.22436026222 /// MAE: 198.06114187529502 /// Mean CRPS: 6350.295679644809
basismodel_ts(96)_states(15) --- loglike-Train: -153312.1905280291 /// loglike-Test: -164392.88949816892 /// MAE: 205.35362505357585 /// Mean CRPS: 6644.010615892532
basismodel_ts(96)_states(20) --- loglike-Train: -143169.269441539 /// loglike-Test: -152224.87374905334 /// MAE: 201.64080571113496 /// Mean CRPS: 6455.777293943534
basismodel_ts(96)_states(25) --- loglike-Train: -139531.37425629803 /// loglike-Test: -151509.4487744804 /// MAE: 197.79903583903896 /// Mean CRPS: 6410.152493169399
basismodel_ts(96)_states(30) --- loglike-Train: -135908.5067868718 /// loglike-Test: -147915.65734920134 /// MAE: 202.13995792509772 /// Mean CRPS: 6518.4126252276865
basismodel_ts(96)_states(40) --- loglike-Train: -121242.718372745 /// loglike-Test: -132364.29163408134 /// MAE: 196.2666990989635 /// Mean CRPS: 6293.871556238616
basismodel_ts(96)_states(50) --- loglike-Train: -113322.48761581356 /// loglike-Test: -124996.32825612897 /// MAE: 184.49969604832995 /// Mean CRPS: 5927.717753870674
basismodel_ts(96)_states(60) --- loglike-Train: -107835.41065783244 /// loglike-Test: -120324.28515902758 /// MAE: 190.88988482346764 /// Mean CRPS: 6155.554047131148

Timing: {8410.516}sec

----- Evaluate seasonmodel_hh(5) -----

seasonmodel_(5)_spring_states(5) --- loglike-Train: -8902.54197468696 /// loglike-Test: -10533.390227759237 /// MAE: 113.07466522441635 /// Mean CRPS: 3828.9949048913045
seasonmodel_(5)_spring_states(10) --- loglike-Train: -8179.151121633222 /// loglike-Test: -9684.997387124913 /// MAE: 102.29309849581185 /// Mean CRPS: 3520.6990489130435
seasonmodel_(5)_spring_states(15) --- loglike-Train: -7904.4958315667245 /// loglike-Test: -9252.926681749112 /// MAE: 101.38748223395049 /// Mean CRPS: 3368.8222373188405
seasonmodel_(5)_spring_states(20) --- loglike-Train: -7787.90816909211 /// loglike-Test: -9326.38357895029 /// MAE: 102.30855969079215 /// Mean CRPS: 3376.2672101449275
seasonmodel_(5)_spring_states(25) --- loglike-Train: -7752.506999597587 /// loglike-Test: -9466.346173543254 /// MAE: 100.93937670105775 /// Mean CRPS: 3396.969429347826
seasonmodel_(5)_spring_states(30) --- loglike-Train: -7623.529203004109 /// loglike-Test: -9628.853230141847 /// MAE: 100.76464253102237 /// Mean CRPS: 3433.785099637681
seasonmodel_(5)_spring_states(40) --- loglike-Train: -7663.098526721951 /// loglike-Test: -9926.682696632328 /// MAE: 99.97482098237431 /// Mean CRPS: 3339.1975769927535
seasonmodel_(5)_spring_states(50) --- loglike-Train: -7334.935721695779 /// loglike-Test: -9678.13280753077 /// MAE: 98.15836940677227 /// Mean CRPS: 3369.2454710114493
seasonmodel_(5)_spring_states(60) --- loglike-Train: -7312.553952274999 /// loglike-Test: -10464.545168405311 /// MAE: 100.82314716284268 /// Mean CRPS: 3467.5366847826085

seasonmodel_(5)_summer_states(5) --- loglike-Train: -6358.919081462256 /// loglike-Test: -7570.526159016593 /// MAE: 79.54628049996683 /// Mean CRPS: 2629.6016757246375
seasonmodel_(5)_summer_states(10) --- loglike-Train: -5934.739437189905 /// loglike-Test: -7304.827460527857 /// MAE: 73.9308307885329 /// Mean CRPS: 2507.995810688406
seasonmodel_(5)_summer_states(15) --- loglike-Train: -5756.233230676933 /// loglike-Test: -7255.197730220856 /// MAE: 76.06303406917313 /// Mean CRPS: 2538.7843070652175
seasonmodel_(5)_summer_states(20) --- loglike-Train: -5673.002882561343 /// loglike-Test: -7281.073605514144 /// MAE: 74.0127249665469 /// Mean CRPS: 2503.6465126811595
seasonmodel_(5)_summer_states(25) --- loglike-Train: -5503.329592808337 /// loglike-Test: -7307.182022511675 /// MAE: 74.72095534949757 /// Mean CRPS: 2515.083106884058
seasonmodel_(5)_summer_states(30) --- loglike-Train: -5943.331702275358 /// loglike-Test: -7392.690928547633 /// MAE: 78.58856025371786 /// Mean CRPS: 2572.1013360507245
seasonmodel_(5)_summer_states(40) --- loglike-Train: -5658.513793331963 /// loglike-Test: -7392.920429124997 /// MAE: 75.54175804710073 /// Mean CRPS: 2523.4895833333335
seasonmodel_(5)_summer_states(50) --- loglike-Train: -5248.423639478155 /// loglike-Test: -7502.434240017847 /// MAE: 75.57339952571844 /// Mean CRPS: 2518.1426630434785
seasonmodel_(5)_summer_states(60) --- loglike-Train: -5439.427030580315 /// loglike-Test: -7527.663772468628 /// MAE: 75.92292145962449 /// Mean CRPS: 2539.084125905797

seasonmodel_(5)_fall_states(5) --- loglike-Train: -9158.027967937755 /// loglike-Test: -10688.820290454598 /// MAE: 120.45814598371226 /// Mean CRPS: 3945.470924908425
seasonmodel_(5)_fall_states(10) --- loglike-Train: -8415.474959519624 /// loglike-Test: -9986.83148439015 /// MAE: 107.46656049598386 /// Mean CRPS: 3576.145032051282
seasonmodel_(5)_fall_states(15) --- loglike-Train: -8182.054270656118 /// loglike-Test: -10075.555771847392 /// MAE: 105.72859351828102 /// Mean CRPS: 3524.347870879121
seasonmodel_(5)_fall_states(20) --- loglike-Train: -7706.5738226354315 /// loglike-Test: -9862.995829085696 /// MAE: 107.25582092195889 /// Mean CRPS: 3534.727335164835
seasonmodel_(5)_fall_states(25) --- loglike-Train: -7757.017551694695 /// loglike-Test: -9840.098350899441 /// MAE: 105.39529295722784 /// Mean CRPS: 3522.5943223432223
seasonmodel_(5)_fall_states(30) --- loglike-Train: -7847.669097162085 /// loglike-Test: -10014.541125597469 /// MAE: 105.35283984956016 /// Mean CRPS: 3524.318108974359
seasonmodel_(5)_fall_states(40) --- loglike-Train: -7517.042406499732 /// loglike-Test: -9878.123266752293 /// MAE: 104.13734396250436 /// Mean CRPS: 3484.681662087912
seasonmodel_(5)_fall_states(50) --- loglike-Train: -7274.71932918256 /// loglike-Test: -10296.772677413443 /// MAE: 105.15917596343276 /// Mean CRPS: 3546.4349816849817
seasonmodel_(5)_fall_states(60) --- loglike-Train: -7172.582748154759 /// loglike-Test: -10212.71560763854 /// MAE: 108.02870822088171 /// Mean CRPS: 3613.799679478478

seasonmodel_(5)_winter_states(5) --- loglike-Train: -15333.728410594533 /// loglike-Test: -19028.18923661335 /// MAE: 286.11233922504664 /// Mean CRPS: 9777.952266483517
seasonmodel_(5)_winter_states(10) --- loglike-Train: -12324.468932780155 /// loglike-Test: -13855.08708346246 /// MAE: 121.3748272436593 /// Mean CRPS: 4031.685554029304
seasonmodel_(5)_winter_states(15) --- loglike-Train: -11652.66186890705 /// loglike-Test: -13583.99551494906 /// MAE: 119.4171183163274 /// Mean CRPS: 3934.655792124542
seasonmodel_(5)_winter_states(20) --- loglike-Train: -11447.030280938392 /// loglike-Test: -13063.631366261536 /// MAE: 115.68419744583352 /// Mean CRPS: 3847.290064102564
seasonmodel_(5)_winter_states(25) --- loglike-Train: -11363.578017341504 /// loglike-Test: -13445.93280524057 /// MAE: 117.05235436397369 /// Mean CRPS: 3918.9246784871797
seasonmodel_(5)_winter_states(30) --- loglike-Train: -11241.685969632126 /// loglike-Test: -13412.85329696049 /// MAE: 118.28333307405426 /// Mean CRPS: 3935.2410714285716
seasonmodel_(5)_winter_states(40) --- loglike-Train: -11116.106169092 /// loglike-Test: -13396.293097655654 /// MAE: 116.71031053021711 /// Mean CRPS: 3879.42090214652
seasonmodel_(5)_winter_states(50) --- loglike-Train: -10874.150925200755 /// loglike-Test: -13427.68293466361 /// MAE: 115.89112060306195 /// Mean CRPS: 3921.978021978022
seasonmodel_(5)_winter_states(60) --- loglike-Train: -10729.822979784345 /// loglike-Test: -14074.495283787439 /// MAE: 121.61291467310116 /// Mean CRPS: 4025.252893772893